

Figure S1 / Table S4 — APGI Twelve-Prediction Matrix by Tier
(Paper 4, supplementary; cross-referenced from §7.4)
Canonical equation: $S_t = \Pi^e |z^e| + \Pi_{\text{eff}}^i |z^i| > \theta_t$

	Prediction	Protocol	Effect target	Falsification	Horizon	Status
Tier 3 Computational	P1: HEP amplitude predicts P3b trial-by-trial (Π^l proxy)	EEG cardiac-phase	$r \geq 0.35$	HEP-P3b $r < 0.10$	Now	Partial
	P2: Detection rate higher at diastole vs. systole	EEG cardiac-phase	$d' \geq 0.5$	Null cardiac-phase effect	Now	Pending
	P3: Somatic-marker agent outperforms β -lesion under volatility	Agent simulation	$\eta^2 \geq 0.25$	No reward advantage	Now	Confirmed
	P4: vmPFC-insula PPI higher in anticipation > outcome	fMRI PPI	$\beta \geq 0.20$	PPI null in anticipation	Now	Pending
Tier 2 Info-Theoretic	P5: High-gamma bimodal at frontoparietal threshold	iEEG	$BF \geq 10$	Unimodal gamma	2-5 yr	Pending
	P6: pIC TMS abolishes HEP-PCI coupling	TMS-EEG	$\Delta PCI \geq 0.10$	No TMS dissociation	2-5 yr	Pending
	P7: Masking effectiveness $\propto p_{crit}$ (not stimulus energy)	Visual masking LNN	$AUC \geq 0.80$	p_{crit} null predictor	2-5 yr	Speculative
	P8: 1/f exponent inflections at L1-L4 maturation ages	Developmental EEG	H inflection $\Delta \geq 0.05$	Absent/out-of-sequence	2-5 yr	Speculative
Tier 1 Thermodynamic	P9: κ (ATP/bit) within 10^{13} - 10^{14} range	^{31}P -MRS + iEEG	95% CI overlaps range	κ outside range	5-10 yr	Speculative
	P10: BOLD-calibrated fMRI matches double-bridge estimate	Calibrated fMRI	$r^2 \geq 0.60$	Systematic deviation	5-10 yr	Speculative
	P11: Bistability preserved across 94% of parameter space	Monte Carlo	94.2%	<80% bistability	Now	Confirmed
	P12: Post-block θ elevation in DoC patients	DoC iEEG	Cohen's $d \geq 0.8$	No threshold re-elevation	5-10 yr	Pending

Validation status

- Confirmed
- Partial
- Pending
- Speculative